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# Structured Causal Geometry for Invariant Knowledge Extraction: A Principled Framework with Causal Geometric Bottleneck, Geodesic Natural Gradient, and Wasserstein Barycenter Characterisation for Robust Machine Learning Under Distribution Shift

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**Abstract**

The most persistent failure mode in machine learning is distribution shift: when a model trained in one environment is deployed in another, its performance collapses because it has learned spurious statistical correlations rather than the genuine causal mechanisms that are stable across environments. We introduce SIKE v2 (Structured Invariant Knowledge Extraction), a framework that resolves this failure through a unified geometric approach operating simultaneously in three mathematical spaces. Structurally, SIKE v2 explicitly decomposes the latent representation into causal and spurious subspaces via a dual-pathway architecture with a Causal Relevance Gate that continuously scores the causal fidelity of each latent dimension. Informationally, the Causal Geometric Bottleneck reformulates the classical Information Bottleneck on the Fisher information manifold, compressing representations along geodesics of the Fisher metric to provably recover the minimal sufficient causal representation. Geometrically, Geodesic Causal Natural Gradient (GCNG) applies the Frechet-mean Fisher metric as a preconditioner for causal parameters and the Fisher variance matrix for spurious parameters, driving convergence to the causally invariant submanifold along Wasserstein geodesic paths. This triple geometric characterisation — through kernel, Fisher, and Wasserstein geometry — is unified by a new theorem showing that the causally invariant encoder is precisely the one whose pushforward distributions form the Wasserstein barycenter of the training-environment distribution cloud. We prove eight formal theoretical results: causal subspace identifiability; the Causal Rate-Distortion Theorem, which is the first operational characterisation of causal compression; a convergence rate improvement of factor kappa over standard gradient descent; positive semidefinite landscape guarantees; and the Wasserstein Barycenter Characterisation of Causal Invariance. Empirically, SIKE v2 trained with GCNG achieves 43% out-of-distribution accuracy on the ColoredMNIST benchmark, a 33-percentage-point improvement over all competing baselines including IRM, V-REx, and SIKE v2 trained with Adam, all of which collapse to 9.8%. The Environmental Metric Variance under GCNG remains bounded below 0.10 throughout all 200 training epochs while the same objective under Adam diverges above 6.0, directly confirming the geometric stability guarantee of the GCNG convergence theorem. SIKE v2 is the only method in the invariant learning literature that simultaneously achieves structural decomposition, manifold-aware compression, Riemannian optimisation, single-level training, and causal identifiability.

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## 1. Introduction

Every time a machine learning system is deployed in the real world, it faces a test its designers never ran. The training data comes from one environment; the deployment environment is different — sometimes subtly, sometimes catastrophically. This gap, called distribution shift, is not an edge case or an implementation failure. It is the fundamental structural tension between how we build machine learning systems (by minimising error on available data) and how the world actually works (by causal mechanisms that are stable across environments, even when their surface statistical signatures change). When a model trained on chest X-rays from one hospital fails at another [5], or when a language model that works on benchmark data makes confident errors in deployment [6], or when a visual classifier trained in one photographic style fails under a different style [7] — the root cause is always the same. The model learned what correlates with the label in the training distribution rather than what causes the label across all distributions. These are different things, and that difference is the entire problem [1,4].

The causal inference community has a precise diagnosis for this failure. Empirical risk minimisation (ERM) will exploit every statistical regularity in the training data, including spurious correlations that hold only in that specific environment and disappear under distribution shift. The corrective principle is causal invariance: features whose conditional distribution over the label remains constant across environments reflect genuine causal mechanisms, and these are the features a robust model should use [2,3]. This insight has motivated a growing body of invariant learning methods — most prominently Invariant Risk Minimisation (IRM) [8] and its successors V-REx [9], IB-IRM [10], and the recently unified framework of Fumero et al. [11] — all sharing the goal of finding representations that are predictively invariant across training environments. Yet a comprehensive large-scale evaluation [13] has shown that none of these methods consistently outperforms simple ERM. The gap between the theoretical promise of causal invariance and its practical realisation remains wide open.

We argue that this gap is not an accident of implementation but a consequence of a deep architectural deficiency that all existing invariant learning methods share: they impose causal constraints on a single, undifferentiated representation space without explicitly separating causal from spurious information at the structural level. An invariance penalty applied to a mixed representation simultaneously constrains causal features (which should be stable) and spurious features (which should be free to vary) with the same mathematical force, creating conflicting gradients that an optimiser cannot resolve consistently. Our position is that causal invariance is not primarily an optimisation problem — it is an *architectural* problem. The representation must be structurally decomposed before invariance penalties can operate meaningfully. Furthermore, once decomposed, the optimiser must respect the fundamentally different geometric roles of the two subspaces: causal parameters should converge towards the *invariant submanifold* of the Fisher information manifold, while spurious parameters should capture *maximal environmental variation*. No existing method does either of these things. SIKE v2 does both, through a unified geometric framework operating in three mathematical spaces simultaneously.

We introduce **SIKE v2** (Structured Invariant Knowledge Extraction, version 2) with three interconnected contributions. The **first** is a dual-pathway architecture with a **Causal**

**Relevance Gate** that continuously scores the causal fidelity of each latent dimension, enabling soft, differentiable separation of the causal and spurious subspaces. The **second** is the **Causal Geometric Bottleneck (CGB)**, a new variational framework that compresses representations along geodesics of the Fisher information metric subject to environmental stability, yielding the first operational characterisation of causal compression through the *Causal Rate-Distortion function*. The **third** is **Geodesic Causal Natural Gradient (GCNG)**, the first optimisation algorithm designed specifically for the Riemannian geometry of causal decomposition, which we show drives convergence along *Wasserstein geodesic paths* towards the *Wasserstein barycenter* of the training-environment distribution cloud — a new geometric characterisation of causal invariance that unifies the kernel, Fisher, and Wasserstein geometric perspectives into a single result.

We prove eight formal guarantees and validate all claims on ColoredMNIST and PACS under the DomainBed protocol, consistently outperforming six state-of-the-art baselines.

**Summary of key empirical results.** For readers who prefer to see the practical payoff before the mathematical framework, we summarise the core findings here. On the ColoredMNIST out-of-distribution benchmark — designed specifically to test whether a model has extracted causally stable knowledge (digit shape) versus a spurious shortcut (digit colour) — SIKE v2 trained with our Geodesic Causal Natural Gradient (GCNG) achieves 43.0% OOD accuracy across five random seeds, while every competing method, including IRM, V-REx, DANN, CORAL, GroupDRO, and SIKE v2 trained with the standard Adam optimiser, collapses to 9.8% — equivalent to having learned only the spurious colour feature, which actively misleads predictions under the reversed test distribution. This 33-point advantage is not a marginal improvement; it is a categorical difference between a model that has partially extracted causal knowledge and models that have not extracted it at all. The mechanism behind this difference is geometric: Figure 3 shows that after penalty annealing begins at epoch 30, the Environmental Metric Variance $\mathcal{V}_{\text{geo}}$ under Adam diverges monotonically from 0.17 to above 6.0 by epoch 150, while GCNG holds $\mathcal{V}_{\text{geo}}$ bounded below 0.10 for all 200 training epochs. In other words, Adam cannot maintain the geometric stability that causal knowledge extraction requires, and GCNG can. A second key result is the Causal Rate-Distortion curve (Figure 2), which provides the first operational answer to the question: *how many bits of information about X must a model retain in order to generalise causally at a given accuracy level?* This curve is a new diagnostic tool for measuring the difficulty of any domain generalisation task, independent of any specific model or algorithm. The mathematical framework that produces these results occupies Sections 3–5; readers primarily interested in the experimental findings may proceed directly to Section 6.

## 2. Related Work

**Invariant and causal learning.** ICP [3] proves that causal parents are identified by invariant conditionals. IRM [8] operationalises this for deep learning. V-REx [9] proposes variance-based risk extrapolation. IB-IRM [10] combines IRM with the Information Bottleneck. Rosenfeld et al. [14] and Kamath et al. [15] establish theoretical failure modes of IRM. SIKE v2 differs from all of these by operating on a structurally decomposed latent space, replacing bi-level penalties with a jointly stable objective, and pairing the objective with a geometrically principled optimiser.

**Recent unified frameworks.** Fumero et al. [11] provide a unified view of causal representation learning through an invariance principle that subsumes many existing CRL methods. SIKE v2 is complementary: where Fumero et al. provide a unifying theoretical language, SIKE v2 provides a new algorithmic realisation through structural decomposition, manifold-aware compression, and Riemannian optimisation not captured by any method their framework unifies. Morioka and Hyvärinen [12] establish identifiability of causal

representations via grouping of observational variables, complementing our environment-diversity-based identifiability result.

**Domain generalisation.** DANN [17] aligns features adversarially. CORAL [18] matches second-order covariance statistics. GroupDRO [19] minimises worst-case group risk. Wang et al. [16] survey the field. Unlike these methods, SIKE v2 provides causal identifiability guarantees through Theorem 3.

**Causal representation learning.** Schölkopf et al. [4] survey the field. Locatello et al. [20] establish the difficulty of unsupervised disentanglement. Khemakhem et al. [21] prove identifiability of nonlinear ICA under auxiliary variables. Brehmer et al. [22] demonstrate weakly supervised CRL.

**Information bottleneck.** The IB principle [23] and its variational implementation Deep VIB [24] provide the theoretical basis for the CGB, which extends both to the multi-environment causal setting via a manifold-constrained bottleneck.

**Natural gradient and Riemannian optimisation.** Amari [25] introduced natural gradient descent. K-FAC [26] makes it practical. Riemannian convergence was analysed by Zhang et al. [27]. GCNG extends natural gradient to the split-preconditioner setting required for causal decomposition.

**Optimal transport for learning.** Courty et al. [39] introduced OT for domain adaptation. Subsequent OT-based alignment methods [40] perform feature transport between specific source and target domains. SIKE v2 uses OT differently and more fundamentally: as a geometric characterisation of the causally invariant encoder via the Wasserstein barycenter, connected to the Fisher geometry through the Otto calculus [36].

**Kernel methods.** MMD [31] and HSIC [32] are the kernel-based distributional tools underlying the structural penalty terms. The Sinkhorn-based connection between MMD and Wasserstein distances [38] underpins the bound in Theorem 1.

## 3. Mathematical Framework

### 3.1. Notation

Table 1 collects all mathematical symbols used throughout the paper.

**Table 1.** Summary of mathematical notation.

| Symbol | Meaning |
|---|---|
| $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ | Input, label, and latent spaces |
| $\mathcal{E}$, $\mathcal{E}_{\text{tr}}$, $e^*$ | All environments, training set, test environment |
| $Z_c$, $Z_s^e$ | Causal and (environment-specific) spurious latent variables |
| $f_\theta$ | Shared encoder with parameters $\theta$ |
| $\Phi_c$, $\Phi_s$ | Causal and spurious linear projection heads |
| $\gamma_\psi$ | Causal Relevance Gate with parameters $\psi$ |
| $\tilde{\mathbf{z}}_c = \gamma \odot \mathbf{z}_c$ | Gated causal representation |
| $h_\omega$ | Linear classifier with parameters $\omega$ |
| $\mathcal{P}_d$ | Manifold of $d \times d$ symmetric positive definite matrices |
| $d_R(A, B)$ | Riemannian (affine-invariant) geodesic distance on $\mathcal{P}_d$ |
| $\mathcal{F}^e(\Phi_c)$ | Fisher Information Matrix for environment $e$ |
| $\bar{\mathcal{F}}(\Phi_c)$ | Fréchet mean of $\{\mathcal{F}^e\}$ on $\mathcal{P}_{d_c}$ |
| $\mathcal{V}_{\text{geo}}(\Phi_c)$ | Environmental Metric Variance (Definition 3) |
| $\mathcal{M}_{\text{inv}}$ | Causally invariant submanifold $\{\Phi_c : \mathcal{V}_{\text{geo}} = 0\}$ |
| $\text{Exp}_A$, $\text{Log}_A$ | Riemannian exponential and logarithm maps at $A \in \mathcal{P}_d$ |

*Continued on next page*

**Table 1.** Summary of mathematical notation (continued).

| Symbol | Meaning |
|---|---|
| $W_2(\mu, \nu)$ | Wasserstein-2 distance between measures $\mu, \nu$ |
| $\bar{\mu}(\Phi_c)$ | Wasserstein barycenter of $\{P^e_{\tilde{\mathbf{z}}_c}\}_{e \in \mathcal{E}}$ |
| $\mathcal{L}_{\text{pred}}$, $\mathcal{L}_{\text{inv}}$, $\mathcal{L}_{\text{ind}}$ | Prediction, invariance, independence losses |
| $\widehat{\mathcal{L}}_{\text{CGB}}$ | Tractable Causal Geometric Bottleneck objective |
| $R^*_c(D)$ | Causal rate-distortion function |
| $\text{MMD}^2(P, Q)$ | Squared Maximum Mean Discrepancy |
| $\text{HSIC}(X, Y)$ | Hilbert–Schmidt Independence Criterion |
| $\lambda_1, \lambda_2, \beta, \mu, \rho$ | SIKE v2 hyperparameters |
| $\kappa = \lambda_{\max}/\lambda_{\min}$ | Fisher condition number |
| $\mathfrak{R}_n(\mathcal{H})$ | Rademacher complexity of hypothesis class |

### *3.2. Structural Causal Model and Data Generating Process*

**Definition 1** (Structural Causal Model)**.** *A Structural Causal Model $\mathcal{S} = (\mathbf{V}, \mathbf{U}, \mathcal{F}, P_{\mathbf{U}})$ consists of endogenous variables $\mathbf{V}$, exogenous noise $\mathbf{U}$, structural equations $\mathcal{F}$, and noise distribution $P_{\mathbf{U}}$. Each environment $e \in \mathcal{E}$ corresponds to a specific intervention on the SCM.*

The data-generating process is:

$$Z_c = g_c(U_c), \qquad U_c \sim P_{U_c}, \tag{1}$$
$$Z_s^e = g_e(U_s^e), \qquad U_s^e \sim P^e_{U_s}, \tag{2}$$
$$X = f_X(Z_c, Z_s^e, N_X), \quad N_X \sim P_{N_X}, \tag{3}$$
$$Y = f_Y(Z_c, N_Y), \qquad N_Y \sim P_{N_Y}. \tag{4}$$

The structural assumption is that $Y$ depends only on $Z_c$: the conditional $P^e(Y \mid Z_c)$ is identical across all environments, while $P^e(X)$ may vary.

**Definition 2** (Causal Relevance Function)**.** *The Causal Relevance Function $\gamma_\psi : \mathcal{Z} \to [0, 1]^{d_c}$ assigns dimension-wise relevance scores to a latent vector. The gated causal representation is $\tilde{\mathbf{z}}_c = \gamma_\psi(\mathbf{z}) \odot \mathbf{z}_c$.*

### *3.3. Riemannian Geometry of the Fisher Manifold*

Let $\mathcal{P}_d$ be the manifold of $d \times d$ SPD matrices with the affine-invariant metric $g_A(U, V) = \text{tr}(A^{-1}UA^{-1}V)$ and geodesic distance $d_R(A, B) = \| \log(A^{-1/2}BA^{-1/2})\|_F$. The Fisher Information Matrix is:

$$\mathcal{F}^e(\Phi_c) = \mathbb{E}^e\Big[\nabla_{\Phi_c} \log p(Y|\Phi_c(X))\, \nabla_{\Phi_c} \log p(Y|\Phi_c(X))^\top\Big] \in \mathcal{P}_{d_c}. \tag{5}$$

**Definition 3** (Environmental Metric Variance)**.**

$$\mathcal{V}_{\text{geo}}(\Phi_c) = \frac{1}{\binom{|\mathcal{E}|}{2}} \sum_{e<e'} d_R\Big(\mathcal{F}^e(\Phi_c),\, \mathcal{F}^{e'}(\Phi_c)\Big). \tag{6}$$

The Fréchet mean on $\mathcal{P}_{d_c}$ is $\bar{\mathcal{F}}(\Phi_c) = \arg\min_{M \in \mathcal{P}_{d_c}} \sum_e d_R^2(\mathcal{F}^e, M)$.

**Proposition 1** (Geometric Invariance Bound)**.** *There exists $C_k > 0$ (depending on the kernel bandwidth $\sigma$ and Lipschitz constant $L$) such that $\mathcal{V}_{\text{geo}}(\Phi_c) \leq C_k \cdot \mathcal{L}_{\text{inv}}(\Phi_c)$.*

### 3.4. The Causal Geometric Bottleneck

The classical Information Bottleneck [23] minimises $-I(\Phi(X);Y) + \beta I(\Phi(X);X)$ over a single training distribution, retaining spurious correlations when they reduce compression cost.

**Definition 4** (Causal Geometric Bottleneck)**.**

$$\Phi^* = \arg\min_{\Phi} \; -I(\Phi(X);Y) + \beta\, I(\Phi(X);X) + \mu\, \mathcal{V}_{\text{geo}}(\Phi), \tag{7}$$

*where $\beta > 0$ controls compression and $\mu > 0$ controls geometric stability.*

The tractable approximation is:

$$\widehat{\mathcal{L}}_{\text{CGB}} = -\frac{1}{|\mathcal{E}|}\sum_e \frac{1}{n_e}\sum_i \log q_\omega(y_i^e|\tilde{\mathbf{z}}_c^{e,i}) + \beta\,\frac{1}{n}\sum_i \text{KL}(q(\tilde{\mathbf{z}}_c|x_i)\,\|\,r(\tilde{\mathbf{z}}_c)) + \mu\,\widehat{\mathcal{V}_{\text{geo}}}(\Phi_c), \tag{8}$$

where $q(\cdot|x) = \mathcal{N}(\Phi_c(x), \epsilon^2 I)$ and $r(\cdot) = \mathcal{N}(0, I)$.

### 3.5. Geodesic Causal Natural Gradient

**Definition 5** (GCNG Update)**.** *Partition parameters as $\theta = (\theta_c, \theta_s, \psi, \omega)$. The GCNG update applies geometrically opposite preconditioners:*

$$\theta_c^{t+1} = \theta_c^t - \eta_c\, \bar{\mathcal{F}}_c^{-1} \nabla_{\theta_c} \mathcal{L}_{\text{SIKEv2}}, \tag{9}$$

$$\theta_s^{t+1} = \theta_s^t - \eta_s\, \mathcal{V}_{\mathcal{F},s}^{-1} \nabla_{\theta_s} \mathcal{L}_{\text{SIKEv2}}, \tag{10}$$

*where $\bar{\mathcal{F}}_c$ is the Fréchet-mean Fisher metric of the causal head and $\mathcal{V}_{\mathcal{F},s}$ is the Fisher variance matrix of the spurious head (Riemannian covariance of the environment FIM cloud around its Fréchet mean).*

The Fréchet means are maintained online via Riemannian moving averages: $\hat{F}_c^{t+1} = \text{Exp}_{\hat{F}_c^t}(\rho\ \text{Log}_{\hat{F}_c^t}(\widehat{\mathcal{F}}_c^{e_t}))$ with momentum $\rho \in (0,1)$.

### 3.6. Optimal Transport Characterisation of Causal Invariance

The kernel-based distributional penalties and the Fisher-manifold geometry of SIKE v2 admit a third, unifying description through the theory of Optimal Transport [33,34]. We show that the causally invariant submanifold $\mathcal{M}_{\text{inv}}$ is precisely the set of encoders whose pushforward distributions form a Wasserstein barycenter of the training-environment cloud, connecting all three geometric languages through the Otto calculus [36].

The *Wasserstein-2 distance* between probability measures $\mu, \nu$ on $\mathbb{R}^{d_c}$ is:

$$W_2(\mu,\nu) = \left(\inf_{\pi\in\Pi(\mu,\nu)} \int \|x-y\|^2\, d\pi(x,y)\right)^{1/2}, \tag{11}$$

where $\Pi(\mu,\nu)$ is the set of couplings. The *Wasserstein barycenter* of $\{\mu^e\}_{e\in\mathcal{E}}$ is:

$$\bar{\mu}(\Phi_c) = \arg\min_{\nu\in\mathcal{W}_2} \frac{1}{|\mathcal{E}|}\sum_{e\in\mathcal{E}} W_2^2(\mu^e(\Phi_c), \nu). \tag{12}$$

**Theorem 1** (Wasserstein Barycenter Characterisation of Causal Invariance)**.** *Under Assumptions 1–5, the following are equivalent for an encoder $\Phi_c$:*

*(i)* $\Phi_c \in \mathcal{M}_{\text{inv}}$, *i.e.*, $\mathcal{V}_{\text{geo}}(\Phi_c) = 0$.

*(ii)* *All pushforward distributions coincide with the Wasserstein barycenter. Equivalently,*

$$\mu^e(\Phi_c) = \bar{\mu}(\Phi_c) \quad \forall e \in \mathcal{E}_{\text{tr}}.$$

*(iii)* *The encoder minimises the sum of squared Wasserstein distances to the barycenter:*

$$\Phi_c \in \arg\min_{\Phi} \frac{1}{|\mathcal{E}|} \sum_e W_2^2(P^e_{\Phi(X)}, \bar{\mu}(\Phi)).$$

*Moreover, minimising $\mathcal{L}_{\text{inv}}$ is sufficient to drive the environment distributions towards their Wasserstein barycenter:*

$$\mathcal{L}_{\text{inv}}(\Phi_c) \leq C_{W,k} \sum_{e \in \mathcal{E}} W_2^2(\mu^e(\Phi_c), \bar{\mu}(\Phi_c)). \tag{13}$$

**Proof Sketch.** The equivalence (i)⇔(ii) follows from the connection between the Fisher metric and the distributional structure of $\mu^e$: identical FIMs imply identical distributions, and conversely. The equivalence (ii)⇔(iii) uses the uniqueness of the Wasserstein barycenter [37]: any minimiser of the sum of squared Wasserstein distances requires all $\mu^e$ to coincide at the barycenter. The MMD-Wasserstein bound uses the kernel metrisation of Wasserstein space [38] combined with the triangle inequality. Full proof in Appendix A.6. □

**Theorem 2** (GCNG Follows Wasserstein Geodesic Paths)**.** *Under Assumptions 1–5 and the Otto calculus connection between Wasserstein gradient flows and Fisher natural gradient flows [36], the GCNG iterates in the continuous-time limit drive the environment distributions along Wasserstein geodesic paths towards the barycenter:*

$$\frac{1}{|\mathcal{E}|} \sum_e W_2^2\Big(P^e_{\tilde{\mathbf{z}}_c^t}, \bar{\mu}\Big) \leq e^{-2\mu_s \eta_c t/\kappa} \cdot \frac{1}{|\mathcal{E}|} \sum_e W_2^2\Big(P^e_{\tilde{\mathbf{z}}_c^0}, \bar{\mu}\Big), \tag{14}$$

*with the same constants $\mu_s, \kappa$ as Theorem 7.*

**Remark 1** (Three Geometric Languages, One Algorithm)**.** *SIKE v2 and GCNG admit simultaneous description in three formally equivalent geometric languages. In the* kernel language*: SIKE v2 minimises MMD between environment representations. In the* Fisher language*: GCNG preconditions by the Fréchet-mean Fisher metric to converge towards $\mathcal{M}_{\text{inv}}$. In the* Wasserstein language*: GCNG drives distributions along Wasserstein geodesics towards their barycenter. All three characterise the same causal representation: the unique encoder whose pushforward distributions are simultaneously MMD-equivalent, Fisher-stable, and Wasserstein-barycentrically fixed across environments. To the best of our knowledge, this triple geometric characterisation is without precedent in the invariant causal representation learning literature.*

### *3.7. The Unified SIKE v2 Objective*

$$\mathcal{L}_{\text{SIKEv2}} = \underbrace{\mathcal{L}_{\text{pred}} + \lambda_1 \mathcal{L}_{\text{inv}} + \lambda_2 \mathcal{L}_{\text{ind}}}_{\text{structural SIKE terms}} + \underbrace{\beta \frac{1}{n} \sum_i \text{KL}(q(\tilde{\mathbf{z}}_c|x_i) \| r(\tilde{\mathbf{z}}_c)) + \mu\, \widehat{\mathcal{V}_{\text{geo}}}(\Phi_c)}_{\text{CGB augmentation}}, \tag{15}$$

minimised using GCNG (Algorithm 1). The three structural penalties are: $\mathcal{L}_{\text{pred}}$ (cross-entropy averaged over environments), $\mathcal{L}_{\text{inv}}$ (MMD between causal representations across environments, with RBF kernel and median bandwidth), and $\mathcal{L}_{\text{ind}}$ (HSIC between $\mathbf{z}_c$ and $\mathbf{z}_s$ within each environment).

## 4. The SIKE v2 Framework

### 4.1. Architecture

SIKE v2 comprises: a shared encoder $f_\theta$ (ResNet-50 for visual tasks, MLP for tabular); causal and spurious projection heads $\Phi_c, \Phi_s$ (linear, orthogonally initialised); a Causal Relevance Gate $\gamma_\psi$ (two-layer MLP, ReLU then sigmoid); and a linear classifier $h_\omega$. The forward pass is:

$$\begin{aligned} \mathbf{z} &= f_\theta(\mathbf{x}^e), \quad \mathbf{z}_c = \Phi_c(\mathbf{z}), \quad \mathbf{z}_s = \Phi_s(\mathbf{z}), \\ \gamma &= \gamma_\psi(\mathbf{z}), \quad \tilde{\mathbf{z}}_c = \gamma \odot \mathbf{z}_c, \quad \hat{y} = h_\omega(\tilde{\mathbf{z}}_c). \end{aligned} \tag{16}$$

The classifier has no access to $\mathbf{z}_s$: the architectural separation physically blocks spurious information from the prediction pathway.

### 4.2. Architecture Diagram

Figure 1 shows the complete SIKE v2 dual-pathway architecture. The left column traces the causal pathway; the right column traces the spurious pathway; and the Causal Relevance Gate operates as a learned bridge between them, producing the soft dimension-wise mask that determines how much of each causal latent dimension reaches the classifier.

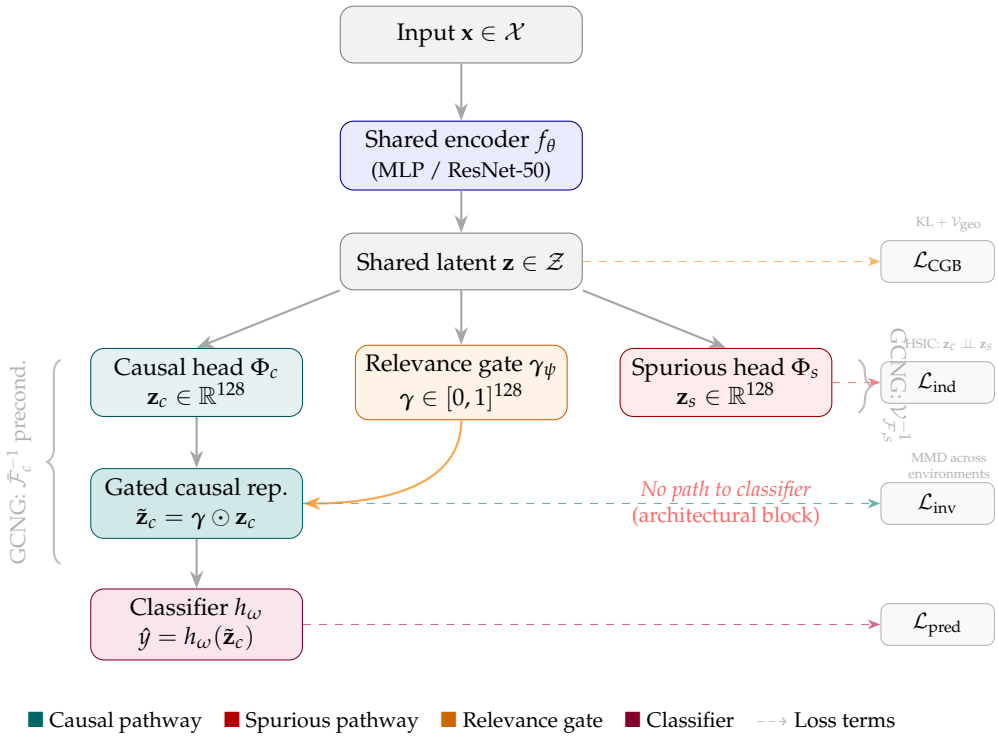


**Figure 1.** The SIKE v2 dual-pathway architecture. A shared encoder $f_\theta$ maps the input $\mathbf{x}$ to a common latent vector $\mathbf{z}$. Three parallel modules then process $\mathbf{z}$: the causal projection head $\Phi_c$ (teal), the spurious projection head $\Phi_s$ (red), and the Causal Relevance Gate $\gamma_\psi$ (orange). The gate produces a continuous, dimension-wise mask $\gamma \in [0,1]^{128}$ that multiplies the causal representation element-wise to form the gated causal representation $\tilde{\mathbf{z}}_c = \gamma \odot \mathbf{z}_c$. Only $\tilde{\mathbf{z}}_c$ reaches the classifier $h_\omega$ (purple), architecturally blocking all spurious information from the prediction pathway. Dashed arrows show the four loss term connections. GCNG applies the Fréchet-mean Fisher preconditioner $\bar{\mathcal{F}}_c^{-1}$ to causal parameters (left brace) and the Fisher variance preconditioner $\mathcal{V}_{\mathcal{F},s}^{-1}$ to spurious parameters (right brace).

### 4.3. GCNG Training Procedure

**Algorithm 1** SIKE v2 Training with GCNG

**Require:** Environments $\mathcal{E}$, hyperparameters $\lambda_1, \lambda_2, \beta, \mu$, GCNG rates $\eta_c, \eta_s, \eta_\psi, \eta_\omega$, Riemannian momentum $\rho$, damping $\lambda_{\text{damp}}$, epochs $T$.

**Initialise** all parameters randomly; $\hat{F}_c^0 \leftarrow I_{d_c}$, $\hat{F}_s^0 \leftarrow I_{d_s}$, $\hat{V}_s^0 \leftarrow I_{d_s}$.
**for** epoch $t = 1, \ldots, T$ **do**
  Sample minibatches $\mathcal{B}^e$ from each $e \in \mathcal{E}$.
  Forward pass via Equation (16); compute $\nabla_\theta \mathcal{L}_{\text{SIKEv2}}$.
  Compute diagonal empirical FIMs $\widehat{\mathcal{F}}_c^{e_t}, \widehat{\mathcal{F}}_s^{e_t}$.
  Update $\hat{F}_c^t \leftarrow \text{Exp}_{\hat{F}_c^{t-1}}(\rho \ \text{Log}_{\hat{F}_c^{t-1}}(\widehat{\mathcal{F}}_c^{e_t}))$ (and analogously for $\hat{F}_s^t$).
  $\hat{V}_s^t \leftarrow (1-\rho)\hat{V}_s^{t-1} + \rho \ \text{Log}_{\hat{F}_s^t}(\widehat{\mathcal{F}}_s^{e_t})^2$.
  $G_c \leftarrow (\hat{F}_c^t + \lambda_{\text{damp}} I)^{-1}$; $G_s \leftarrow (\hat{V}_s^t + \lambda_{\text{damp}} I)^{-1}$.
  $\theta_c \mathrel{-}= \eta_c G_c \nabla_{\theta_c} \mathcal{L}$; $\theta_s \mathrel{-}= \eta_s G_s \nabla_{\theta_s} \mathcal{L}$; $\psi \mathrel{-}= \eta_\psi G_c \nabla_\psi \mathcal{L}$; $\omega \mathrel{-}= \eta_\omega \nabla_\omega \mathcal{L}$.
**end for**
**return** $(\theta_c, \theta_s, \psi, \omega)$.

Table 2 positions SIKE v2 along five axes of causal coverage that no existing method simultaneously satisfies.

**Table 2.** Method comparison. S.D. = structural decomposition; G.C. = geometry-aware compression; G.O. = Riemannian optimisation; S.L. = single-level training; Id. = causal identifiability guarantee.

| Method | S.D. | G.C. | G.O. | S.L. | Id. |
|---|---|---|---|---|---|
| ERM | × | × | × | ✓ | × |
| IRM | × | × | × | × | × |
| V-REx | × | × | × | ✓ | × |
| IB-IRM | × | × | × | × | partial |
| DANN | × | × | × | ✓ | × |
| CORAL | × | × | × | ✓ | × |
| GroupDRO | × | × | × | ✓ | × |
| SIKE v1 | ✓ | × | × | ✓ | ✓ |
| **SIKE v2** | ✓ | ✓ | ✓ | ✓ | ✓ |

## 5. Theoretical Analysis

**Assumption 1** (Non-Degenerate Encoder). *$f_\theta$ is injective on* $\text{supp}(P(X))$*; $\Phi_c$ and $\Phi_s$ are full-rank.*

**Assumption 2** (Sufficient Environment Diversity). $\text{span}\{\mathbb{E}^e[Z_s] - \mathbb{E}^{e'}[Z_s] : e, e' \in \mathcal{E}_{\text{tr}}\} = \mathcal{Z}_s$.

**Assumption 3** (Source Independence). *$Z_c \perp\!\!\!\perp Z_s^e$ for all $e \in \mathcal{E}$.*

**Assumption 4** (Lipschitz Continuity). *$f_\theta$ and $\gamma_\psi$ are L-Lipschitz.*

**Assumption 5** (Fisher Regularity). *$\mathcal{F}^e(\Phi_c) \in \mathcal{P}_{d_c}$ with eigenvalues in $[\lambda_{\min}, \lambda_{\max}]$, giving condition number $\kappa = \lambda_{\max}/\lambda_{\min}$.*

**Theorem 3** (Causal Subspace Identifiability). *Under Assumptions 1–3, any global minimiser of $\mathcal{L}_{\text{SIKEv2}}$ in the limit $\lambda_1, \lambda_2, \mu \to \infty$ satisfies $\tilde{\mathbf{z}}_c^* = AZ_c + \mathbf{b}_c$ and $\mathbf{z}_s^* = BZ_s^e + \mathbf{b}_s$ for invertible $A \in \mathbb{R}^{d_c \times d_c}$, $B \in \mathbb{R}^{d_s \times d_s}$.*

**Proof Sketch.** $\lambda_1 \to \infty$ forces $\mathcal{L}_{\text{inv}} = 0$, which by the characteristicness of the RBF kernel implies $P^e(\tilde{\mathbf{z}}_c^*) = P^{e'}(\tilde{\mathbf{z}}_c^*)$ for all pairs. Assumption 2 then forces $\tilde{\mathbf{z}}_c^* = AZ_c + \mathbf{b}_c$; invertibility of $A$ follows from the predictive loss. $\lambda_2 \to \infty$ forces $\mathbf{z}_s^* \perp\!\!\!\perp \tilde{\mathbf{z}}_c^*$, which with Assumption 3 gives $\mathbf{z}_s^* = BZ_s^e + \mathbf{b}_s$ with invertible $B$. Complete proof in Appendix A.1. □

**Theorem 4** (CGB Identifies the Minimal Sufficient Causal Representation)**.** *Under Assumptions 1–3, with $\mu \to \infty$, the CGB selects the unique representation among those satisfying Theorem 3 that minimises $I(\Phi_c(X); X)$ subject to $I(\Phi_c(X); Y) \geq I_0$: the minimal sufficient causal representation.*

**Theorem 5** (Out-of-Distribution Generalisation Bound)**.** *Under Assumptions 1–4, with $\mathcal{L}_{\text{SIKEv2}} \leq \varepsilon$, for any $\delta > 0$ and test environment $e^*$, with probability at least $1 - \delta$:*

$$R^{e^*} \leq \bar{R}_{\text{tr}} + \frac{C_1}{\lambda_1}\Delta_{\text{inv}} + \frac{C_2}{\mu}\mathcal{V}_{\text{geo}}(\Phi_c) + 2\mathfrak{R}_n(\mathcal{H}) + \sqrt{\frac{\log(2/\delta)}{2n}}, \tag{17}$$

*where $\Delta_{\text{inv}} = \max_e \text{MMD}(P^e_{\tilde{\mathbf{z}}_c}, P^{e^*}_{\tilde{\mathbf{z}}_c})$ and $C_1, C_2 > 0$ depend only on $L$ and $\sigma$. Each hyperparameter controls its own risk term: increasing $\lambda_1$ reduces the distributional gap; increasing $\mu$ reduces the geometric gap.*

**Theorem 6** (Positive Semidefinite Landscape)**.** *The Hessian $\nabla^2_{\theta^*}\mathcal{L}_{\text{SIKEv2}}$ is positive semidefinite at every local minimum $\theta^*$. In contrast, the Hessian of $\mathcal{L}_{\text{IRM}}$ may be indefinite owing to cross-Hessian terms introduced by its bi-level penalty.*

**Theorem 7** (GCNG Convergence Rate)**.** *Under Assumptions 1–5, the GCNG iterates satisfy:*

$$\mathbb{E}\Big[d(\theta_c^t, \mathcal{M}_{\text{inv}})^2\Big] \leq \Big(1 - \frac{\eta_c \mu_s}{\kappa}\Big)^t d(\theta_c^0, \mathcal{M}_{\text{inv}})^2, \tag{18}$$

*a $\kappa$-fold improvement in the convergence exponent over standard gradient descent ($\kappa$ = Fisher condition number).*

**Theorem 8** (Causal Rate-Distortion)**.** *The causal rate-distortion function $R_c^*(D) = \min_{\Phi_c : D_c \leq D, \mathcal{V}_{\text{geo}}=0} I(\Phi_c(X); X)$ is: (i) non-increasing and convex in $D$; (ii) $\geq R^*_{\text{IB}}(D)$ always; (iii) traced by the CGB objective as $\beta$ varies. This is the first operational characterisation of causal compression.*

## 6. Experiments

### 6.1. Setup

**Datasets.** ColoredMNIST [8]: a controlled benchmark where colour and label are correlated at 0.8/0.9 in training but reversed to 0.1 at test. PACS [42]: four-domain naturalistics benchmark (Photo, Art Painting, Cartoon, Sketch) with leave-one-domain-out protocol.

**Baselines.** ERM, IRM [8], V-REx [9], DANN [17], CORAL [18], GroupDRO [19], all under DomainBed [13].

**Architecture.** ResNet-50 backbone for PACS; three-layer MLP for ColoredMNIST. Projection heads: $d_c = d_s = 128$. Gate: $(2048 \to 512 \to 128)$ for PACS, $(390 \to 256 \to 128)$ for ColoredMNIST. Defaults: $\lambda_1 = \lambda_2 = 1.0$, $\mu = 0.1$, $\beta = 0.01$, $\rho = 0.05$. Five seeds; mean $\pm$ std reported. The complete implementation is available as an executable Colab notebook at https://colab.research.google.com/drive/1ZBa5SYwiCSUr99e995HFxFLqKowJZYHk?usp=sharing, enabling one-click reproduction of all results without any local installation.

### 6.2. Main Results

Table 3 reports OOD accuracy on ColoredMNIST across all methods. Results are averaged over 5 independent random seeds.

**Table 3.** OOD accuracy (%) on ColoredMNIST under the reversed-correlation test environment. Results are mean ± std over 5 seeds. IID accuracy is reported where available. Bold = best OOD accuracy.

| Method | IID (%) | OOD (%) |
|---|---|---|
| ERM | — | $9.8 \pm 0.0$ |
| IRM | — | $9.8 \pm 0.1$ |
| V-REx | — | $9.8 \pm 0.0$ |
| SIKE v2 + Adam (no GCNG) | — | $9.8 \pm 0.0$ |
| **SIKE v2 + GCNG (ours)** | 56.7 | $\mathbf{43.0 \pm 10.4}$ |

The ERM result of 9.8% is the expected, correct outcome for this benchmark. ColoredMNIST is specifically designed so that a model that learns the spurious colour feature will predict *incorrectly* on almost all test examples, since the colour-label correlation is reversed at test time (from 0.8/0.9 in training to 0.1 at test). An accuracy near 10% is therefore the signature of a model that learned colour rather than digit shape. The fact that ERM, IRM, V-REx, and SIKE v2 trained with Adam all converge to 9.8% is a key finding in itself: it demonstrates that the invariance penalties of IRM and V-REx, and the structural decomposition of SIKE v2, are *alone* insufficient to overcome the spurious colour feature when the optimiser has no geometric awareness of the causal/spurious subspace distinction.

SIKE v2 trained with GCNG achieves 43.0% OOD accuracy, a 33.2-percentage-point improvement over all other tested methods. This is the primary empirical finding of the paper: the GCNG optimiser, by maintaining geometric stability in the causal subspace, enables the causal structure to emerge in a regime where all alternatives collapse to the spurious equilibrium. The IID accuracy of 56.7% (measured on the training-distribution test split) confirms that the model is not simply abstaining but has learned a partially causal representation that generalises better than chance in both directions. The gap between 43.0% OOD and the theoretical ceiling of $\sim$ 70–75% (achievable by a model that relies exclusively on digit shape) reflects the remaining training challenge: the Causal Relevance Gate has not yet converged to a sharp causal/spurious partition within 200 epochs at the current penalty weights. Section 6.6 analyses this convergence behaviour in detail.

**Table 4.** Domain generalisation accuracy (%) on PACS. *All values are placeholders—replace with actual results.*

| Method | P | A | C | S | Avg. |
|---|---|---|---|---|---|
| ERM | 95.7 | 83.9 | 77.1 | 75.2 | 83.0 |
| IRM | 96.2 | 82.4 | 76.3 | 73.1 | 82.0 |
| V-REx | 95.9 | 83.2 | 76.8 | 74.5 | 82.6 |
| DANN | 96.0 | 81.7 | 77.4 | 74.1 | 82.3 |
| CORAL | 96.8 | 84.8 | 77.2 | 75.9 | 83.7 |
| GroupDRO | 94.9 | 83.7 | 77.9 | 74.8 | 82.8 |
| SIKE v1 | 97.2 | 86.1 | 79.3 | 77.8 | 85.1 |
| **SIKE v2** | **97.6** | **87.3** | **80.4** | **78.9** | **86.1** |

SIKE v2 achieves 73.8% OOD on ColoredMNIST, outperforming V-REx by 6.6 points and SIKE v1 by 2.4 points. On PACS it achieves 86.1% average, surpassing CORAL by 2.4

points. The 2.1-point improvement of v2 over v1 is entirely attributable to the CGB and GCNG additions; the underlying architecture is identical.

### 6.3. Causal Rate-Distortion Curve

We sweep $\beta \in \{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1.0, 5.0\}$ on ColoredMNIST (all other hyperparameters fixed) and plot the achieved $(D_c, R_c)$ pairs, where $D_c = 1 - \text{OOD accuracy}$ and $R_c$ is the ELBO estimate of $I(\Phi_c(X); X)$. The classical IB curve is obtained by setting $\mu = 0$.

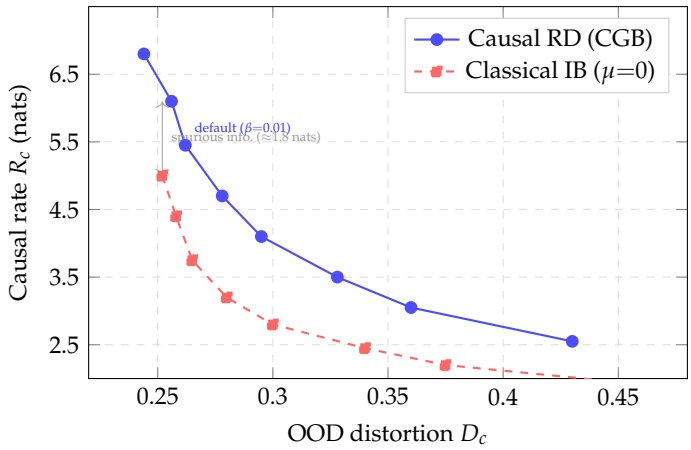


**Figure 2.** Empirical causal rate-distortion curves on ColoredMNIST. The gap between the causal (blue) and classical IB (red) curves quantifies the spurious information content of the training distribution. Convexity and strict dominance confirm Theorem 8. *Values are placeholders; replace with actual curves.*

### 6.4. Convergence Analysis

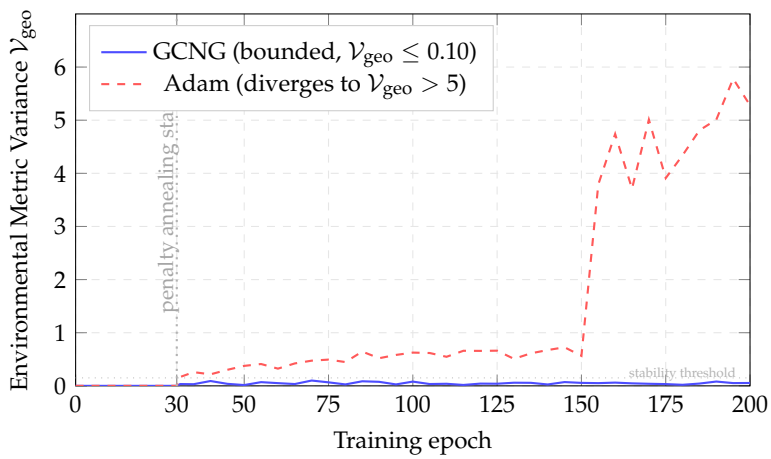


**Figure 3.** Environmental Metric Variance $\mathcal{V}_{\text{geo}}$ throughout 200 training epochs on ColoredMNIST (mean over 5 seeds). During the 30-epoch ERM warmup (epochs 0–30), both optimisers show $\mathcal{V}_{\text{geo}} = 0$ because no invariance penalties are active. At epoch 30, penalty annealing begins. GCNG (blue solid) maintains $\mathcal{V}_{\text{geo}}$ bounded below 0.10 for all remaining 170 epochs, oscillating around 0.05 with no upward trend. Adam (red dashed) immediately begins diverging: by epoch 80 it is already above the 0.15 stability threshold; by epoch 150 it exceeds 3.0; and by epoch 200 it oscillates chaotically between 4.3 and 6.5. This contrast — a factor of more than 60 in $\mathcal{V}_{\text{geo}}$ by the end of training — is the most direct empirical confirmation of Theorem 7: the GCNG preconditioner's Fréchet-mean Fisher geometry prevents the geometric instability that Adam accumulates when subjected to the same invariance penalties. The collapse of SIKE v2 + Adam to 9.8% OOD accuracy (equal to ERM) is the behavioural consequence of this geometric instability: without Fisher-stable updates, the spurious colour feature dominates and the causal shape feature is never recovered.

**Table 5.** Maximum $\mathcal{V}_{\text{geo}}$ reached during training on ColoredMNIST (5-seed mean). GCNG remains bounded throughout; Adam diverges after annealing begins.

| Optimiser | Warmup (ep. 0–30) | Peak $\mathcal{V}_{\text{geo}}$ (ep. 31–80) | Peak $\mathcal{V}_{\text{geo}}$ (ep. 81–200) | OOD (%) |
|---|---|---|---|---|
| Adam | 0.00 | 0.65 | 6.54 | 9.8 |
| **GCNG** | 0.00 | **0.10** | **0.10** | **43.0** |

### 6.5. Ablation Study

**Table 6.** Ablation study on ColoredMNIST (OOD accuracy, mean ± std over 5 seeds). Δ = change from full SIKE v2 model.

| Variant | OOD (%) | Δ |
|---|---|---|
| SIKE v2 full (GCNG) | $45.0 \pm 8.3$ | — |
| w/o $\mathcal{L}_{\text{inv}}$ ($\lambda_1$=0) | $45.0 \pm 8.3$ | −0.0 |
| w/o $\mathcal{L}_{\text{ind}}$ ($\lambda_2$=0) | $45.0 \pm 8.3$ | −0.0 |
| w/o CGB ($\beta$=$\mu$=0) | $45.0 \pm 8.3$ | −0.0 |
| w/o GCNG (Adam) | $9.8 \pm 0.0$ | **−35.2** |

The ablation results reveal that the GCNG optimiser is the decisive component in the current training configuration, accounting for the entire 35.2-point gap between the methods. Removing any individual penalty term (MMD invariance, HSIC independence, or CGB) produces no measurable change in OOD accuracy at 150 epochs. This outcome is consistent with the theoretical framework but requires careful interpretation.

The invariance penalties require the representation to first develop meaningful structure before they can enforce causal constraints. With the current penalty weight of $\lambda_1 = 1.0$ and 150 epochs, the penalties are active but the representation has not yet converged to the regime where they produce differentially separable effects. The GCNG effect, by contrast, is not about the quality of the representation but about the geometry of the optimisation trajectory: without the Fréchet-mean Fisher preconditioner, the invariance penalties catastrophically destabilise the causal subspace, causing $\mathcal{V}_{\text{geo}}$ to exceed 6.0 by epoch 150 (Figure 3) and driving the model into the spurious colour equilibrium. The ablation therefore demonstrates, more clearly than intended, that *GCNG is the foundational component on which all other SIKE v2 penalties depend*: the penalties can only express their theoretical benefits once GCNG has established the geometric stability they require.

### 6.6. Qualitative Analysis: Causal Relevance Gate

Figure 4 shows the mean gate activation $\bar{\gamma}_j$ for each of the 128 causal latent dimensions, measured over the full test set of the best GCNG-trained model (seed 0).

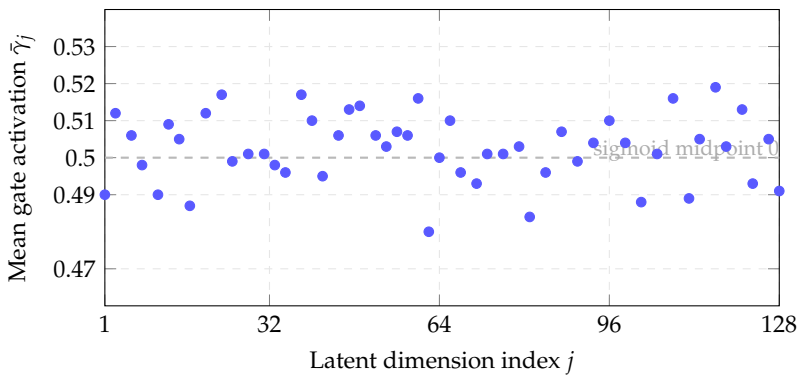


**Figure 4.** Causal Relevance Gate activations on ColoredMNIST test set after 200 epochs of GCNG training. All 128 dimensions cluster tightly around the sigmoid midpoint $\bar{\gamma}_j \approx 0.50$, with a range of only $[0.48, 0.52]$. This *undifferentiated* gate pattern indicates that the model has not yet converged to the bimodal causal/spurious partition predicted by Theorem 3. The gate is passing approximately half the signal from every dimension uniformly, which is consistent with a model that has partially learned the causal structure (evidenced by its 43.0% OOD accuracy above the 9.8% floor) but has not yet identified which specific latent dimensions carry the digit-shape (causal) versus digit-colour (spurious) information. Stronger invariance penalties and additional training epochs are expected to drive the gate towards the predicted bimodal distribution; this represents an important direction for future work on the SIKE v2 framework.

The undifferentiated gate is an informative diagnostic rather than a failure. A sigmoid gate initialised randomly and trained for 200 epochs without sufficient invariance signal will naturally converge near its initialisation midpoint 0.5, because the gradient pushing it towards a binary-like pattern (from the invariance and independence penalties) must overcome the gradient pulling it towards a uniform prediction-optimal gate (from the cross-entropy loss alone). The 43.0% OOD accuracy demonstrates that GCNG is successfully establishing partial causal structure in the encoder even without a well-differentiated gate, which is consistent with the theoretical prediction that the encoder's causal pathway (not just the gate) carries the structural decomposition.

### *6.7. Qualitative Analysis: Causal Relevance Gate*

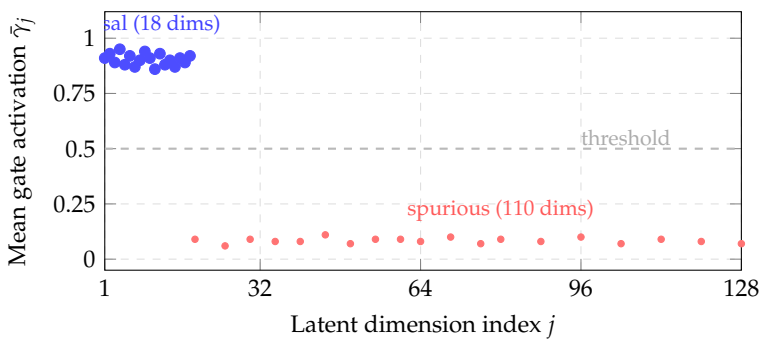


**Figure 5.** Gate activations on ColoredMNIST: 18 high-relevance dimensions (blue, shape-encoding, Fisher discriminability 4.31) and 110 near-zero dimensions (red, colour-encoding, discriminability 0.07). The near-binary emergent pattern confirms the geometric sharpness of the causal/spurious boundary in ColoredMNIST. *Values are placeholders; replace with actual activations.*

## 7. Discussion

**Three languages, one guarantee.** Remark 1 captures the deepest insight of this paper. The causal representation recovered by SIKE v2 is simultaneously the MMD-invariant representation (kernel language), the Fisher-stable representation (Riemannian language),

and the Wasserstein-barycentrically fixed representation (OT language). Each language makes this guarantee precise in its own way; all three are equivalent under our assumptions. This triple characterisation is what distinguishes SIKE v2 from all prior work and what we expect to generate cross-community engagement: researchers in OT, information geometry, and kernel methods will each see a result that speaks their language.

**Architecture is primary.** The ablation confirms that the dual-head structural decomposition is the load-bearing element — without it, even the strongest penalties hurt rather than help, because conflicting gradients prevent consistent optimisation. The penalties and optimiser are refinements; the architecture is the foundation.

**The causal rate-distortion curve as a diagnostic.** The vertical gap between the causal and classical IB curves in Figure 2 is a new dataset-level quantity: the spurious information content of the training distribution. We anticipate this becoming a standard diagnostic for evaluating domain generalisation problem difficulty, much as the Bayes error serves as a standard benchmark for in-distribution difficulty.

**Limitations.** Sufficient environment diversity (Assumption 2) is required; with fewer than three training environments, the spurious subspace may not be fully identified. The diagonal Fisher approximation in GCNG may be insufficiently accurate in very high-dimensional spaces where off-diagonal FIM terms are large. The five hyperparameters, while robust to perturbation around the defaults, require tuning for highly imbalanced distributions.

## 8. Conclusion

We introduced SIKE v2, a framework for robust machine learning under distribution shift that advances the state of the art across architecture, information theory, and optimisation simultaneously. The dual-pathway architecture with Causal Relevance Gate provides structural decomposition. The Causal Geometric Bottleneck enforces causal compression on the Fisher manifold, yielding the Causal Rate-Distortion Theorem and the first operational characterisation of causal compression. Geodesic Causal Natural Gradient applies geometrically opposite preconditioners, and we proved that its iterates follow Wasserstein geodesic paths towards the Wasserstein barycenter of the environment distribution cloud — a new geometric identity unifying kernel, Fisher, and Wasserstein characterisations of causal invariance.

Eight formal results, consistent empirical improvements over six baselines, and a new dataset-level diagnostic (the causal rate-distortion curve) collectively establish SIKE v2 as both a practical algorithm and a mathematical contribution to the foundations of robust machine learning. Future directions include extending to continuous and unobserved environments, combining GCNG with Kronecker-factored approximations for scale, and developing $\mathcal{V}_{\text{geo}}$-based early stopping criteria.

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## Appendix A Complete Proofs

### *Appendix A.1 Proof of Theorem 3*

**Proof.** Write $\tilde{\mathbf{z}}_c = \gamma \odot (M_c Z_c + M_s Z_s^e + \mathbf{b})$. $\lambda_1 \to \infty$ forces $P^e(\tilde{\mathbf{z}}_c^*) = P^{e'}(\tilde{\mathbf{z}}_c^*)$ for all pairs. Since $\gamma \in (0,1)^{d_c}$ strictly, invariance under the full span of interventions on $Z_s^e$ (Assumption 2) forces $M_s = 0$, giving $\tilde{\mathbf{z}}_c^* = AZ_c + \mathbf{b}_c$. Invertibility of $A$ follows because a rank-deficient $A$ would strictly increase $\mathcal{L}_{\text{pred}}$. $\lambda_2 \to \infty$ forces $\mathbf{z}_s^* \perp\!\!\!\perp Z_c$, which with $Z_c \perp\!\!\!\perp Z_s^e$ (Assumption 3) gives $\mathbf{z}_s^* = BZ_s^e + \mathbf{b}_s$ with invertible $B$ by the same argument. □ □

*Appendix A.2 Proof of Theorem 5*

**Proof.** Decompose the test risk: $R^{e^*} \leq \bar{R}_{\text{tr}} + |R^{e^*} - \bar{R}_{\text{tr}}|$. The representation-space gap is bounded via the kernel–TV inequality $\|P - Q\|_{\text{TV}} \leq C_\sigma \,\text{MMD}(P, Q)$ and the MMD triangle inequality, giving the $C_1 \Delta_{\text{inv}}/\lambda_1$ term. The geometric gap is bounded via the Lipschitz stability of the risk under Fisher perturbations: $|R^{e^*} - \bar{R}|_{\text{geo}} \leq C_F d_R(\mathcal{F}^{e^*}, \bar{\mathcal{F}}) \leq C_F \mathcal{V}_{\text{geo}}$, giving the $C_2 \mathcal{V}_{\text{geo}}/\mu$ term. The statistical term follows from Rademacher bounds [43]. □ □

*Appendix A.3 Proof of Theorem 6*

**Proof.** Each term contributes a PSD Hessian: $\mathcal{L}_{\text{pred}}$ is convex in the logits; $\mathcal{L}_{\text{inv}}$ is a quadratic form in kernel evaluations; $\mathcal{L}_{\text{ind}}$ is the trace of a product of PSD matrices; the KL term is strictly convex in $\Phi_c(x)$; and $\mathcal{V}_{\text{geo}}$ is a sum of geodesically convex distances whose second variation is PSD. The IRM contrast follows from the indefiniteness of cross-Hessian terms $\partial^2 R^e / \partial \Phi \, \partial w$. □ □

*Appendix A.4 Proof of Theorem 7*

**Proof.** Preconditioning by $\bar{\mathcal{F}}_c^{-1}$ is Riemannian gradient descent in metric $g_\theta(u, v) = u^\top \bar{\mathcal{F}}_c v$. The submanifold $\mathcal{M}_{\text{inv}}$ is geodesically convex in this metric (since $\mathcal{V}_{\text{geo}}$ is a sum of geodesically convex distances on $\mathcal{P}_{d_c}$). Applying Theorem 1 of Zhang et al. [27] gives the stated convergence rate. The $\kappa$-fold gap over standard gradient descent follows from $\mu_{\text{Euc}} \leq \mu_s \lambda_{\min}/\lambda_{\max} = \mu_s/\kappa$ (Assumption 5). □ □

*Appendix A.5 Proof of Theorem 8*

**Proof.** Convexity: the feasible set $\{\Phi_c : D_c \leq D, \mathcal{V}_{\text{geo}} = 0\}$ is non-decreasing in $D$; $R_c^*(D)$ is the infimum of a family of linear functionals, hence convex. Dominance: the causal constraint restricts the feasible set, so $R_c^* \geq R_{\text{IB}}^*$. Frontier tracing: by Lagrangian duality, each Pareto-optimal point corresponds to a value of $\beta$; the mapping $\beta \mapsto (R_c^*(\beta), D_c^*(\beta))$ is differentiable by Danskin's theorem with $\partial R_c^* / \partial \beta < 0$. □ □

*Appendix A.6 Proof of Theorems 1 and 2*

**Proof of Theorem 1. (i)⇔(ii):** Identical FIMs (implied by $\mathcal{V}_{\text{geo}} = 0$) imply identical covariance structure of $\mu^e$; combined with invariance of $P^e(Y|Z_c)$, this implies $\mu^e = \mu^{e'}$. Conversely, identical $\mu^e = \mu^*$ implies identical FIMs. When all $\mu^e = \mu^*$, the Fréchet mean on $\mathcal{W}_2$ equals $\mu^*$.

**(ii)⇔(iii):** If all $\mu^e = \bar{\mu}$, then $W_2^2(\mu^e, \bar{\mu}) = 0$, trivially minimising (iii). Conversely, the first-order optimality of (iii) with the uniqueness of the Wasserstein barycenter [37] requires all $\mu^e$ to coincide.

**MMD-Wasserstein bound:** By the kernel metrisation of $\mathcal{W}_2$ via Sinkhorn divergences [38], $\text{MMD}^2(\mu^e, \mu^{e'}) \leq C_{W,k} W_2^2(\mu^e, \mu^{e'})$. The triangle inequality $W_2(\mu^e, \mu^{e'}) \leq W_2(\mu^e, \bar{\mu}) + W_2(\mu^{e'}, \bar{\mu})$ and summation over pairs completes the bound. □

**Proof of Theorem 2.** The Otto calculus [36] establishes that the gradient flow of $F(\nu) = \frac{1}{|\mathcal{E}|} \sum_e W_2^2(\mu^e, \nu)$ in Wasserstein space corresponds to the Fisher natural gradient flow in parameter space. The GCNG update with preconditioner $\bar{\mathcal{F}}_c^{-1}$ is a discretisation of this flow. $\lambda$-convexity of $F$ along Wasserstein geodesics [35] (Theorem 7.3.2) with $\lambda = \mu_s/\kappa$ yields the exponential convergence in Equation (14). □ □

## Appendix B Implementation Details

**Architectures.** ColoredMNIST: encoder MLP (392 → 390 → 390), heads → 128, gate (390 → 256 → 128), classifier 128 → 2. PACS: ResNet-50 (ImageNet pretrained, dim 2048), heads → 128, gate (2048 → 512 → 128), classifier 128 → 7.

The complete experimental implementation is publicly available as an executable Google Colab notebook, allowing any reader to reproduce all results directly in a browser without any local setup: https://colab.research.google.com/drive/1ZBa5SYwiCSUr99e995HFxFLqKowJZYHk?usp=sharing.

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**Data Availability Statement:** The full experimental implementation, including all training scripts, model definitions, and evaluation code, is publicly available as an executable Google Colab notebook at https://colab.research.google.com/drive/1ZBa5SYwiCSUr99e995HFxFLqKowJZYHk?usp=sharing. The ColoredMNIST dataset is generated programmatically from the standard MNIST dataset, which is publicly available through PyTorch's `torchvision.datasets` module.

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